

Smart Waste Collection and Recycling Management System using Python

MODULE 1 DICTIONARY OPERATORS

PSEUDOCODE:

INITIALIZE waste_data as EMPTY DICTIONARY

FUNCTION add_waste(waste_type, quantity, category, recyclable):

 IF waste_type is EMPTY THEN

 DISPLAY "Please enter waste type"

 RETURN

 END IF

waste_data[waste_type] = {

 "quantity": quantity,

 "category": category,

 "recyclable": recyclable

}

DISPLAY "Waste record added successfully"

END FUNCTION

FUNCTION display_waste():

 IF length of waste_data = 0 THEN

 DISPLAY "No waste records available"

 RETURN

 END IF

records = EMPTY STRING

FOR each waste_type, details IN waste_data.items():

 ADD waste_type TO records

 ADD quantity from details TO records

 ADD category from details TO records

 ADD recyclable status from details TO records

END FOR

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DISPLAY records

END FUNCTION

FUNCTION search_waste(waste_type):

IF waste_type is EMPTY THEN

DISPLAY "Please enter waste type to search"

RETURN

END IF

IF waste_type EXISTS IN waste_data THEN

details = waste_data[waste_type]

DISPLAY waste_type

DISPLAY details["quantity"]

DISPLAY details["category"]

DISPLAY details["recyclable"]

ELSE

DISPLAY "Waste record not found"

END IF

END FUNCTION

FUNCTION update_waste(waste_type, quantity, category, recyclable):

IF waste_type is EMPTY THEN

DISPLAY "Please enter waste type"

RETURN

END IF

IF waste_type EXISTS IN waste_data THEN

waste_data[waste_type] = {

"quantity": quantity,

"category": category,

"recyclable": recyclable

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```
}  
  
DISPLAY "Waste record updated successfully"  
  
ELSE  
  
DISPLAY "Waste record not found"  
  
END IF  
  
END FUNCTION  
  
FUNCTION delete_waste(waste_type):  
    IF waste_type is EMPTY THEN  
        DISPLAY "Please enter waste type"  
        RETURN  
    END IF  
  
    IF waste_type EXISTS IN waste_data THEN  
        DELETE waste_data[waste_type]  
        DISPLAY "Waste record deleted successfully"  
    ELSE  
        DISPLAY "Waste record not found"  
    END IF  
  
END FUNCTION  
  
END MODULE
```

MODULE 1 – DICTIONARY OPERATIONS

CODE:

```
# Dictionary  
  
waste_data = {}  
  
# ADD WASTE  
  
def add_waste():  
    waste_type = waste_entry.get()
```

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```
quantity = quantity_entry.get()
category = category_entry.get()
recyclable = recyclable_entry.get()

if waste_type == "":
    messagebox.showerror("Error", "Please enter waste type")
    return

waste_data[waste_type] = {
    "quantity": quantity,
    "category": category,
    "recyclable": recyclable
}

messagebox.showinfo(
    "Success",
    "Waste record added successfully"
)

# DISPLAY WASTE
def display_waste():
    if len(waste_data) == 0:
        messagebox.showinfo(
            "Waste Records",
            "No waste records available"
        )
    return

records = ""

for waste_type, details in waste_data.items():
    records += "Waste Type: " + waste_type + "\n"
    records += "Quantity: " + details["quantity"] + " kg\n"
```

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```
records += "Category: " + details["category"] + "\n"

records += "Recyclable: " + details["recyclable"] + "\n"

records += "-----\n"

messagebox.showinfo("Waste Records", records)

# SEARCH WASTE

def search_waste():

    waste_type = waste_entry.get()

    if waste_type == "":

        messagebox.showerror(

            "Error",

            "Please enter waste type to search"

        )

        return

    if waste_type in waste_data:

        details = waste_data[waste_type]

        result = "Waste Type: " + waste_type + "\n"

        result += "Quantity: " + details["quantity"] + " kg\n"

        result += "Category: " + details["category"] + "\n"

        result += "Recyclable: " + details["recyclable"]

        messagebox.showinfo("Search Result", result)

    else:

        messagebox.showerror(

            "Not Found",

            "Waste record not found"

        )

# UPDATE WASTE

def update_waste():
```

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```
waste_type = waste_entry.get()
quantity = quantity_entry.get()
category = category_entry.get()
recyclable = recyclable_entry.get()

if waste_type == "":
    messagebox.showerror(
        "Error",
        "Please enter waste type"
    )
    Return

if waste_type in waste_data:
    waste_data[waste_type] = {
        "quantity": quantity,
        "category": category,
        "recyclable": recyclable
    }

messagebox.showinfo(
    "Success",
    "Waste record updated successfully"
)

else:
    messagebox.showerror(
        "Not Found",
        "Waste record not found"
    )

# DELETE WASTE

def delete_waste():
```

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```
waste_type = waste_entry.get()

if waste_type == "":
    messagebox.showerror(
        "Error",
        "Please enter waste type"
    )

Return:

if waste_type in waste_data:
    del waste_data[waste_type]
messagebox.showinfo(
    "Success",
    "Waste record deleted successfully"
)

else:
    messagebox.showerror(
        "Not Found",
        "Waste record not found"
    )
```

MODULE 2 file handling

PSEUDOCODE:

FUNCTION save_data():

 OPEN "waste_data.txt" IN WRITE MODE

FOR each waste_type, details IN waste_data.items():

 WRITE waste_type, quantity, category

 and recyclable status TO file

 SEPARATE each value using "|"

END FOR

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CLOSE file

DISPLAY "Waste data saved successfully"

END FUNCTION

FUNCTION load_data():

 TRY:

 OPEN "waste_data.txt" IN READ MODE

 FOR each line IN file:

 REMOVE leading/trailing spaces and newline from line

 IF line is EMPTY THEN

 CONTINUE

 END IF

 data = SPLIT line using "|" "

 waste_type = data[0]

 quantity = data[1]

 category = data[2]

 recyclable = data[3]

 waste_data[waste_type] = {

 "quantity": quantity,

 "category": category,

 "recyclable": recyclable

 }

 END FOR

CLOSE file

DISPLAY "Waste data loaded successfully"

CATCH FileNotFoundError:

 DISPLAY "Waste data file not found"

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END TRY

END FUNCTION

END MODULE

MODULE 2 – FILE HANDLING

CODE:

```
# Module 2: File Handling
```

```
# SAVE DATA
```

```
def save_data():
```

```
    with open("waste_data.txt", "w") as file:
```

```
    for waste_type, details in waste_data.items():
```

```
        file.write(
```

```
            waste_type + "|" +
```

```
            details["quantity"] + "|" +
```

```
            details["category"] + "|" +
```

```
            details["recyclable"] + "\n"
```

```
        )
```

```
    messagebox.showinfo(
```

```
        "Success",
```

```
        "Waste data saved successfully"
```

```
    )
```

```
# LOAD DATA
```

```
def load_data():
```

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```
try:
    with open("waste_data.txt", "r") as file:
        for line in file:
            line = line.strip()
            if line == "":
                continue
            data = line.split("|")
            waste_type = data[0]
            quantity = data[1]
            category = data[2]
            recyclable = data[3]
            waste_data[waste_type] = {
                "quantity": quantity,
                "category": category,
                "recyclable": recyclable
            }
            messagebox.showinfo(
                "Success",
                "Waste data loaded successfully"
            )
        except FileNotFoundError:
            messagebox.showerror(
                "Error",
                "Waste data file not found"
            )
```

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Waste Type:

food waste

Quantity (kg):

5

Category:

wet waste

Recyclable (Yes/No):

no

ADD WASTE

DISPLAY WASTE

SEARCH WASTE

UPDATE WASTE

DELETE WASTE

SAVE DATA

LOAD DATA

Search Result

Waste Type: food waste
Quantity: 5 kg
Category: wet waste
Recyclable: no

OK

Waste Type:

food waste

Quantity (kg):

5

Category:

wet waste

Recyclable (Yes/No):

no

ADD WASTE

DISPLAY WASTE

SEARCH WASTE

UPDATE WASTE

DELETE WASTE

SAVE DATA

LOAD DATA

Waste Records

Waste Type: paper
Quantity: 10 kg
Category: dry waste
Recyclable: yes

Waste Type: plastic
Quantity: 10 kg
Category: dry waste
Recyclable: yes

Waste Type: paper
Quantity: 10 kg
Category: dry waste
Recyclable: yes

Waste Type: plastic
Quantity: 5 kg
Category: dry waste
Recyclable: yes

Waste Type: food waste
Quantity: 5 kg
Category: wet waste
Recyclable: no

OK

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Waste Type:

Quantity (kg):

Category:

Recyclable (Yes/No):

Success

Waste record deleted successfully

OK

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Waste Type:

Quantity (kg):

Category:

Recyclable (Yes/No):

Success

Waste record updated successfully

OK